

# Diagnostic report: Mg I

Computed levels: 30

Matched to NIST: 16

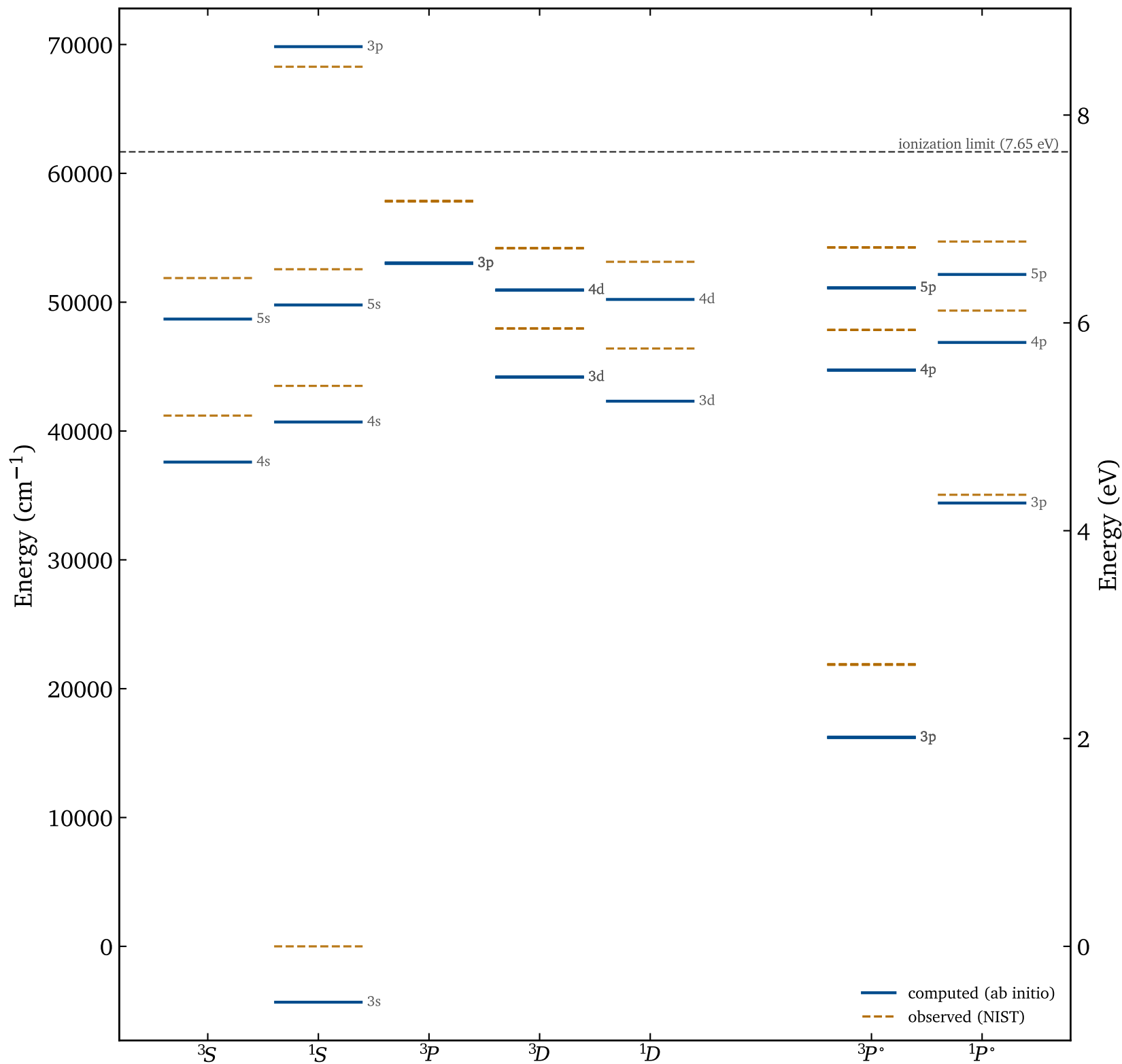
Computed E1 lines: 66

Pre-fit RMS ( $E_{\text{calc}} - E_{\text{obs}}$ ): 29252 cm<sup>-1</sup> (ab initio, no RCE fit yet)

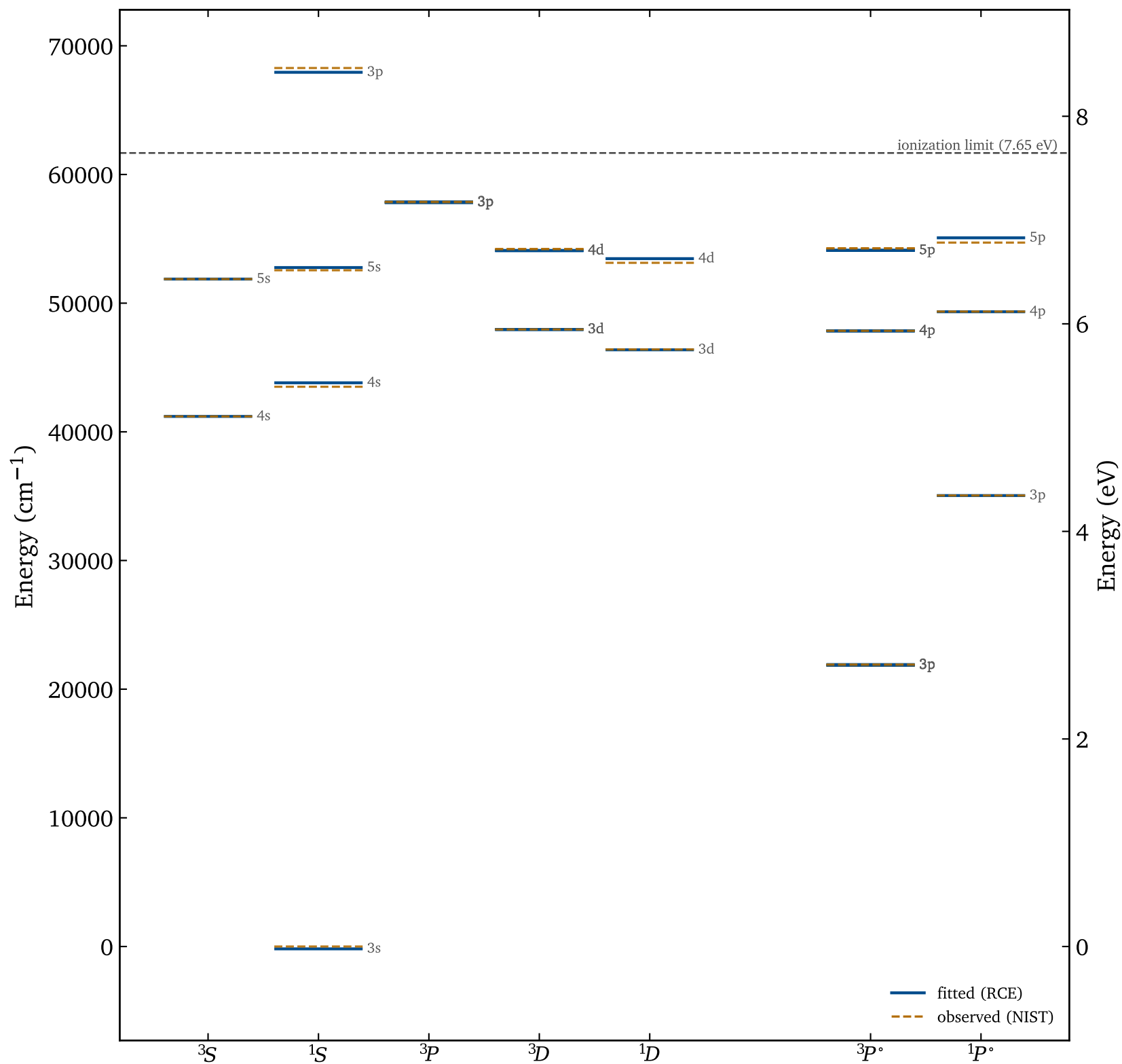
Strongest computed lines:

| lambda(A) | log gf | upper term |
|-----------|--------|------------|
| 3566.85   | +0.789 | (2S) 1P    |
| 29958.38  | +0.521 | (2S) 1P    |
| 41200.71  | +0.509 | (2S) 3P    |
| 2716.41   | +0.470 | (2S) 3P    |
| 16124.69  | +0.438 | (2S) 3P    |
| 51567.09  | +0.429 | (2S) 1P    |
| 13993.58  | +0.406 | (2S) 3P    |
| 3578.48   | +0.365 | (2S) 3P    |

Mg I: term diagram - ab initio vs observed



Mg I: term diagram - fitted vs observed



Mg I: levels — ab initio & fitted vs observed (cm<sup>-1</sup>)

| config | term            | $J$ | $E_{\text{obs}}$ | $E_{\text{abinit}}$ | $\Delta_{\text{abinit}}$ | $E_{\text{fit}}$ | $\Delta_{\text{fit}}$ |
|--------|-----------------|-----|------------------|---------------------|--------------------------|------------------|-----------------------|
| 3s2    | <sup>1</sup> S  | 0   | 0.0              | -4333.0             | -4333.0                  | -181.9           | -181.9                |
| 3s.3p  | <sup>3</sup> P° | 0   | 21850.4          | 16199.0             | -5651.4                  | 21850.2          | -0.2                  |
| 3s.3p  | <sup>3</sup> P° | 1   | 21870.5          | 16213.0             | -5657.5                  | 21870.7          | +0.2                  |
| 3s.3p  | <sup>3</sup> P° | 2   | 21911.2          | 16240.0             | -5671.2                  | 21911.8          | +0.6                  |
| 3s.3p  | <sup>1</sup> P° | 1   | 35051.3          | 34411.0             | -640.3                   | 35047.3          | -4.0                  |
| 3s.4s  | <sup>3</sup> S  | 1   | 41197.4          | 37590.0             | -3607.4                  | 41197.4          | +0.0                  |
| 3s.4s  | <sup>1</sup> S  | 0   | 43503.3          | 40700.0             | -2803.3                  | 43810.7          | +307.4                |
| 3s.3d  | <sup>1</sup> D  | 2   | 46403.1          | 42315.0             | -4088.1                  | 46373.9          | -29.2                 |
| 3s.4p  | <sup>3</sup> P° | 0   | 47841.1          | 44727.0             | -3114.1                  | 47841.0          | -0.1                  |
| 3s.4p  | <sup>3</sup> P° | 1   | 47844.4          | 44730.0             | -3114.4                  | 47844.5          | +0.1                  |
| 3s.4p  | <sup>3</sup> P° | 2   | 47851.2          | 44736.0             | -3115.2                  | 47851.4          | +0.2                  |
| 3s.3d  | <sup>3</sup> D  | 1   | 47957.1          | 44185.0             | -3772.1                  | 47957.1          | +0.0                  |
| 3s.3d  | <sup>3</sup> D  | 2   | 47957.0          | 44185.0             | -3772.0                  | 47957.1          | +0.1                  |
| 3s.3d  | <sup>3</sup> D  | 3   | 47957.0          | 44185.0             | -3772.0                  | 47957.1          | +0.1                  |
| 3s.4p  | <sup>1</sup> P° | 1   | 49346.7          | 46875.0             | -2471.7                  | 49331.6          | -15.1                 |
| 3s.5s  | <sup>3</sup> S  | 1   | 51872.5          | 48690.0             | -3182.5                  | 51872.5          | +0.0                  |
| 3s.5s  | <sup>1</sup> S  | 0   | 52556.2          | 49784.0             | -2772.2                  | 52766.9          | +210.7                |
| 3s.4d  | <sup>1</sup> D  | 2   | 53134.6          | 50213.0             | -2921.6                  | 53460.0          | +325.4                |
| 3s.4d  | <sup>3</sup> D  | 1   | 54192.3          | 50937.0             | -3255.3                  | 54087.2          | -105.1                |
| 3s.4d  | <sup>3</sup> D  | 2   | 54192.3          | 50937.0             | -3255.3                  | 54087.2          | -105.1                |
| 3s.4d  | <sup>3</sup> D  | 3   | 54192.3          | 50937.0             | -3255.3                  | 54087.2          | -105.1                |
| 3s.5p  | <sup>3</sup> P° | 0   | 54248.8          | 51114.0             | -3134.8                  | 54100.5          | -148.3                |
| 3s.5p  | <sup>3</sup> P° | 1   | 54250.1          | 51115.0             | -3135.1                  | 54100.5          | -149.6                |
| 3s.5p  | <sup>3</sup> P° | 2   | 54252.7          | 51118.0             | -3134.7                  | 54100.6          | -152.1                |
| 3s.5p  | <sup>1</sup> P° | 1   | 54706.5          | 52152.0             | -2554.5                  | 55076.7          | +370.2                |
| 3p2    | <sup>3</sup> P  | 0   | 57812.8          | 53006.0             | -4806.8                  | 57811.8          | -1.0                  |
| 3p2    | <sup>3</sup> P  | 1   | 57833.4          | 53022.0             | -4811.4                  | 57831.5          | -1.9                  |
| 3p2    | <sup>3</sup> P  | 2   | 57873.9          | 53054.0             | -4819.9                  | 57870.3          | -3.6                  |
| 3p2    | <sup>1</sup> S  | 0   | 68275.0          | 69834.0             | +1559.0                  | 67941.6          | -333.4                |

